

09f — Database Failure Modes

Key Concepts

Term	Definition
Failure Mode	A way a database can fail or degrade in production
WAL (Write-Ahead Log)	Changes logged to disk before applied — enables crash recovery
Replication Lag	Replica falls behind primary — stale reads possible
Deadlock	Two transactions each wait for a lock the other holds — neither can proceed
Connection Pool	Fixed set of reusable DB connections — can be exhausted under high load
Cascading Failure	One slow query triggers a chain reaction that degrades the whole system

Failure Modes Summary

Failure Mode	Cause	Fix
Hardware failure	Disk/server crash	WAL, replication, backups
Replication lag	Replica behind primary	Route critical reads to primary
Deadlock	Circular lock dependency	Consistent lock ordering; DB auto-detects and kills one transaction
Connection pool exhaustion	Too many concurrent requests	Pool tuning, PgBouncer, query optimization
Data corruption	Crash mid-write, disk error	ACID transactions, backups, point-in-time recovery
Cascading failure	Slow query blocks others → pool fills → timeouts	Query timeouts, circuit breakers, read replicas

Deadlock Flow

Transaction A: locks Row 1 → waits for Row 2

Transaction B: locks Row 2 → waits for Row 1

→ Neither can proceed → DB detects → kills one → it rolls back

Cascading Failure Flow

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Slow query holds locks
→ Other queries pile up waiting
→ Connection pool exhausts
→ New requests time out
→ Entire service degrades
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Industry Examples

- **GitHub (2012)** — replication lag caused stale reads after writes; fixed by routing post-write reads to primary
- **Amazon** — circuit breakers prevent one slow DB from cascading into full service outage
- **Stack Overflow** — connection pool exhaustion during traffic spikes; solved with PgBouncer

Interview Questions

Question	One-Line Answer
What is a deadlock?	Two transactions waiting on each other's locks; DB detects and kills one automatically
What is replication lag?	Replica falls behind primary — reads immediately after a write may return stale data
What is connection pool exhaustion?	All DB connections in use; new requests fail or timeout under high load
What is WAL?	Write-Ahead Log — changes written to log before applied; enables crash recovery
What is a cascading failure?	One slow query causes lock pile-up → connection exhaustion → full service degradation